

How to Install and Use the Project

1. System Requirements

- Python 3.8 or later
- OpenCV
- Ultralytics YOLO
- Webcam (built-in or external)
- A computer with sufficient processing power (GPU recommended for faster performance)

2. Installation Steps

1. **Install Python:** Download and install Python from python.org.
2. **Create a Virtual Environment (Optional but Recommended):**
3. `python -m venv yolo_env`
4. `source yolo_env/bin/activate` # For Linux/macOS
`yolo_env\Scripts\activate` # For Windows

5. **Install Required Libraries:**

```
pip install opencv-python ultralytics
```

3. Running the Project

1. Clone the project repository or download the script.
2. Ensure the YOLO weights file (`yolo8n.pt`) is available in the specified directory.
3. Run the main script:

```
python main.py
```

4. The webcam will activate, and object detection will begin in real-time.
5. Press 'q' to exit the program.

4. Troubleshooting

- **Webcam Not Working?** Ensure it is properly connected and accessible.
- **Missing YOLO Weights?** Download from the official ultralytics repository and place them in the correct directory.
- **Slow Performance?** Use a system with a dedicated GPU for better speed.

By following these steps, users can easily install and run the real-time object detection system. Happy coding!

